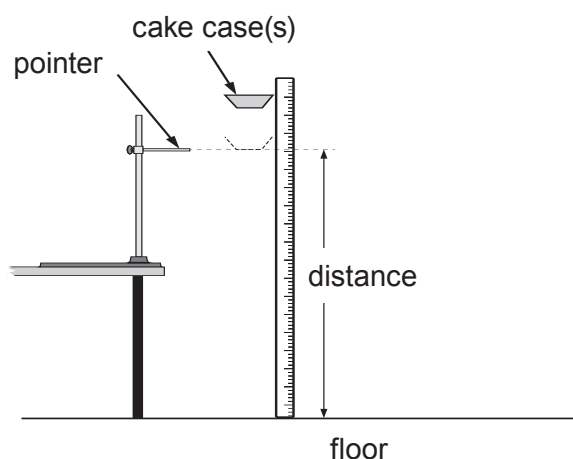


Examiner  
only

- 5.** Students investigate the terminal speed of falling paper cake cases.



- (a) (i) State what is meant by terminal speed.

[1]

- (ii) Describe how you would carry out the investigation to find out how the mass of falling paper cake cases affects their terminal speed. [6 QER]

[6 QER]



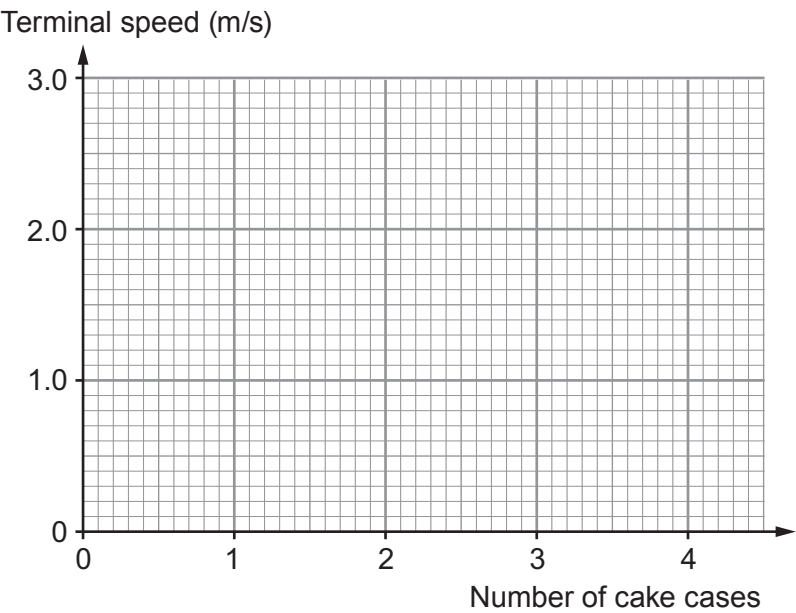
Examiner  
only

(b) The table shows results from the investigation.

| Number of cake cases | Distance (m) | Mean time taken for cake cases to fall 1.50 m (s) | Terminal speed (m/s) |
|----------------------|--------------|---------------------------------------------------|----------------------|
| 0                    |              |                                                   | 0                    |
| 1                    | 1.50         | 0.90                                              | 1.7                  |
| 2                    | 1.50         | 0.68                                              | 2.2                  |
| 3                    | 1.50         | 0.60                                              | .....                |
| 4                    | 1.50         | 0.56                                              | 2.7                  |

(i) **Complete the table** using an equation from page 2 to find the value of the missing terminal speed. *Space for workings.* [2]

(ii) Plot the data in the table on the grid below and draw a suitable line. [3]



(iii) One student suggests that doubling the number of cake cases reduces the **time** to fall by a half. Explain whether the results support this suggestion. [3]

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